



KENYA EDUCATORS PUBLISHERS **KJSEA 2025 BLUE PRINTS**

PREDICTION QUESTIONS & TABLE OF SPECIFICATIONS **INTEGRATED SCIENCE**

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**THIS GUIDE SHOWS YOU WHERE THE KJSEA
(GRADE 9 NATIONAL EXAMS) INTEGRATED SCIENCE
QUESTIONS WILL COME FROM IN THE SYLLABUS
WITH UPTO 80% ACCURACY.**

**THE MAIN OBJECTIVE OF THIS NON-
CONVENTIONAL REVISION GUIDE IS TO HELP
THE CANDIDATE AND THE TEACHER TO SAVE
VALUABLE TIME WHILE REVISING FOR THE
FORTH COMING NATIONAL EXAMS**

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2025 KJSEA Blue Print

INTEGRATED SCIENCE PREDICTION

The purpose of this document is to provide a framework and means of illumination to guide the revision of Integrated Science in preparation of learners for the upcoming national exams.' Therefore, teachers should be keen to look at the KJSEA 2025 KNEC Sample examination.

Confidential!!!

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STRAND 1: SCIENTIFIC INVESTIGATION

Checkpoint 1.1

Grade seven learners in Milimani primary school are investigating the best way to clean up an oil spill in a local pond. One group suggests using only biology based methods, like introducing bacteria that break down oil. Another group proposes using chemistry to create special cleaning agents. How would you explain the importance of combining knowledge from biology, chemistry and physics to develop the most effective and environmentally friendly solution to this problem?

Checkpoint 1.2

You are working in a science laboratory and accidentally knock a chemical container with an acid., The acid spills onto your arm, causing a burn. What should you do to respond effectively to the burn?

Checkpoint 1.3

Marion is preparing for her Chemistry practical examination. She knows that accuracy in measuring liquids is crucial for precise experimental results. She has a variety of equipment in the laboratory, including beakers, test tubes, measuring cylinders and conical flasks. Given that she needs to measure exactly 25 millilitres of hydrochloric acid for her experiment, which piece of glassware equipment should she select to ensure accuracy in her measurement. Why would the other glass apparatus be less suitable for this task?

STRAND 2: MIXTURES, ELEMENTS AND COMPOUNDS

Checkpoint 2.1

Chepkemai was given a task to search for homogeneous mixtures then give example of each. What did she find out?

Checkpoint 2.2

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Ethan is researching on properties of different substances. He has observed that some acidic and basic substances are very useful. Why are some bases and acids important?

Checkpoint 2.3

Grade 8 learners are summarizing their study on elements and compounds. They want to ensure they have covered all important aspects. How can they summarize the relationship between an atom, an element, a molecule and a compound?

Checkpoint 2.4

James is organizing a science fair. He needs to create a display that demonstrates both temporary and permanent changes. Describe two activities or experiments he could use, one for each type of change and explain how they illustrate the characteristics of each change.

Checkpoint 2.5

Tom and Aisha were discussing about the classes of fire and their causes. Name some of the classes of fire and their causes which they could have talked about.

Checkpoint 2.6

Project: Modelling atomic structures

Materials needed

Styrofoam balls (for protons, neutrons and electrons)

Toothpicks or wires (to connect particles)

Markers or paints (to colour code particles)

Clay or playdough (to create nucleus and electron shells)

Printable templates (optional for guiding the structure)

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Checkpoint 2.7

Asili noticed their house had items made from different metals. What are the physical properties of these metals?

Checkpoint 2.8

James and Sofia are discussing the importance of understanding water hardness in their daily lives. They are particularly interested in how it affects their household chores and health.

- (a) Why is it important to appreciate the applications of soft and hard water in everyday life?
- (b) How can understanding water hardness benefit individuals and communities?

STRAND 3: LIVING THINGS AND THEIR ENVIRONMENT

Checkpoint 3.1

How do the male and female reproductive systems work together to enable human reproduction?

Checkpoint 3.2

Julia was asked to describe the role of the excretory system to class. What did she respond?

Checkpoint 3.3

Jane and Martin have been studying cells in their science class. They need to summarize what they have learnt about the structure, function and importance of plant and animal cells. How would they explain their findings to a classmate who missed the lesson?

Checkpoint 3.4

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Your learners are observing two different cells under a microscope: Cell A is in a solution with a high concentration of sugar and Cell B is in a solution with a low concentration of sugar. After some time, you notice changes in both cells.

- (a) What happens to the water movement in each cell?
- (b) How does this demonstrate the process of osmosis?

Checkpoint 3.5

Sarah and Daniel are working on managing menstrual challenges. What strategies should they include in their plan?

Checkpoint 3.6

In your school's garden, you notice two potted plants that are the same - species and size. One plant is placed in a sunny area, while the other is in a shaded part of the garden. After a few weeks, you observe that the plant in the sunny area is growing much better than the plant in the shade.

- (a) How does the difference in light exposure affect the process of photosynthesis in these two plants?
- (b) What might you expect to see in terms of their growth and health?

Checkpoint 3.7

Tom and Aisha are summarizing their study on animal nutrition. They want to ensure they have covered all important aspects. How can they summarize the different modes of nutrition and their significance in animals?

Checkpoint 3.8

Isabel and Noah are working on a project about plant reproduction. They need to explain how different parts of the flower contribute to pollination and fertilization. What steps should they take to ensure they cover all necessary information?

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Checkpoint 3.9

Fatuma and Omar noticed that the local pond has dried up and many animals that depended on it are struggling to survive. How does the lack of water impact the balance between living and non living factors in the environment?

STRAND 4: FORCE AND ENERGY

Checkpoint 4.1

Greenfield School is planning a project to teach learners about electrical energy. The project involves setting up a model home to demonstrate electricity generation, the use of common appliances and safety measures. How would you set up a model home to show the generation, use and safety of electrical energy? Explain the sources, appliances and safety measures and discuss how this helps appreciate the role of electricity in daily life.

Checkpoint 4.2

Joy is curious about why her refrigerator door always stays shut without a latch and why her toy car moves on its own track. Using your knowledge of magnetism, explain how magnets might be responsible for both these -phenomena.

Checkpoint 4.3

Bunyang'anyi village has limited access to electricity- The villagers have decided to harness strong wind in their region to generate electricity for their community. They have installed a windmill connected to a battery system to store excess energy.

- a) Describe the energy transformations that occur from the wind to the powering of a household light bulb in this village.
- b) What safety measures should the villagers practice?

Checkpoint 4.4

Jackson is preparing a presentation on the application of pressure in solids and liquids. What are the applications of pressure in solids and liquids?

Checkpoint 4.5

- a) Mary is setting up a small observatory in her backyard to star gaze at night. She needs a mirror that can gather and focus light from distant stars. What tupe of curved mirror should Mary use for her telescope and why?
- b) Tom is a store owner who wants to improve security in his shop. He needs mirrors that provide a wide field of view so that he can monitor activities in different corners of the store. What type of curved mirror should Tom install in his store and why?

Checkpoint 4.6

There are different categories of waves in the nature. How are waves applied in our day-to-day life?



KJSEA 2025 PREDICTION PAPER



905/1- INTEGRATED SCIENCE (THEORY)

PAPER 1

Section A: Multiple Choice Questions (20 Marks)

1. Which of the following is a non-renewable source of energy?
A) Wind
B) Solar
C) Coal
D) Biomass
2. What is the main gas released during photosynthesis?
A) Carbon dioxide
B) Oxygen
C) Nitrogen
D) Methane
3. Which of the following best describes osmosis?
A) Movement of water from a region of low concentration to high concentration.
B) Movement of water from a region of high concentration to low concentration through a semi-permeable membrane.
C) Movement of solutes from a region of high concentration to low concentration.
D) Movement of gases through a semi-permeable membrane.
4. The smallest unit of an element that retains its chemical properties is:
A) A molecule
B) A compound
C) An atom
D) An ion
5. Which process leads to the formation of clouds?
A) Evaporation
B) Condensation
C) Precipitation
D) Runoff
6. What is the main function of white blood cells in the human body?
A) Transporting oxygen
B) Fighting infections

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- C) Clotting blood
- D) Transporting nutrients
- 7. Which simple machine is used to lift heavy objects with less effort?
 - A) Pulley
 - B) Inclined plane
 - C) Lever
 - D) Screw
- 8. The boiling point of water at sea level is:
 - A) 100°C
 - B) 80°C
 - C) 120°C
 - D) 90°C
- 9. What is the chemical symbol for sodium?
 - A) S
 - B) So
 - C) Na
 - D) N
- 10. Which part of the plant is responsible for absorbing water and nutrients?
 - A) Stem
 - B) Roots
 - C) Leaves
 - D) Flower

Section B: Structured Questions (60 Marks)

1. Introduction to integrated science

a. What is the scientific method?

b. List the steps involved in conducting a scientific investigation.(5 marks)

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2. Measurement and Scientific Units

a. List the basic SI (International System of Units) units for the following quantities: (5 marks)

- i. Length_____.
- ii. Mass._____.
- iii. Time_____.
- iv. Temperature_____.
- v. Volume_____.

• **b.** Convert the following units: (3 marks)

I. 5 meters to centimeters

II. 2.5 kilograms to grams

III. 3.2 liters to milliliters

• **c.** A student measures the mass of a sample and obtains a value of 45.6 grams. What is the precision of this measurement if the instrument has an uncertainty of ± 0.1 g? (2 marks)

3. Matter and Its Properties

• **a.** Define matter and explain the three states of matter (solid, liquid, gas) with examples. (2 marks)

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- **b.** Differentiate between physical and chemical properties of matter with examples. (2 marks)

- **c.** What is the difference between a physical change and a chemical change? Give one example of each. (2 marks)

4. Structure of the Atom

- a.** Draw and label the structure of an atom, including the position of protons, neutrons, and electrons. (2 marks)

b. Define an isotope and provide an example of an isotope of carbon. (2 marks)

c. What is the atomic number and atomic mass of an element? How are they related? (2 marks)

5. The Periodic Table

a. Explain the organization of elements in the periodic table. What information can be obtained from an element's position in the table? (2 marks)

b. What are periods and groups on the periodic table? How are elements in the same group similar? (2 marks)

c. Describe the properties of metals, non-metals, and metalloids. Provide one example of each. (2 marks)



KJSEA 2025 PREDICTION PAPER



905/2- INTEGRATED SCIENCE (PRACTICAL)

PAPER 2

1. **Experiment: Separating a Mixture of Sand and Salt** (20 Marks)

Describe the procedure you would follow to separate a mixture of sand and salt. Include materials and steps.

2. The diagram below shows the water cycle representing various physical changes.

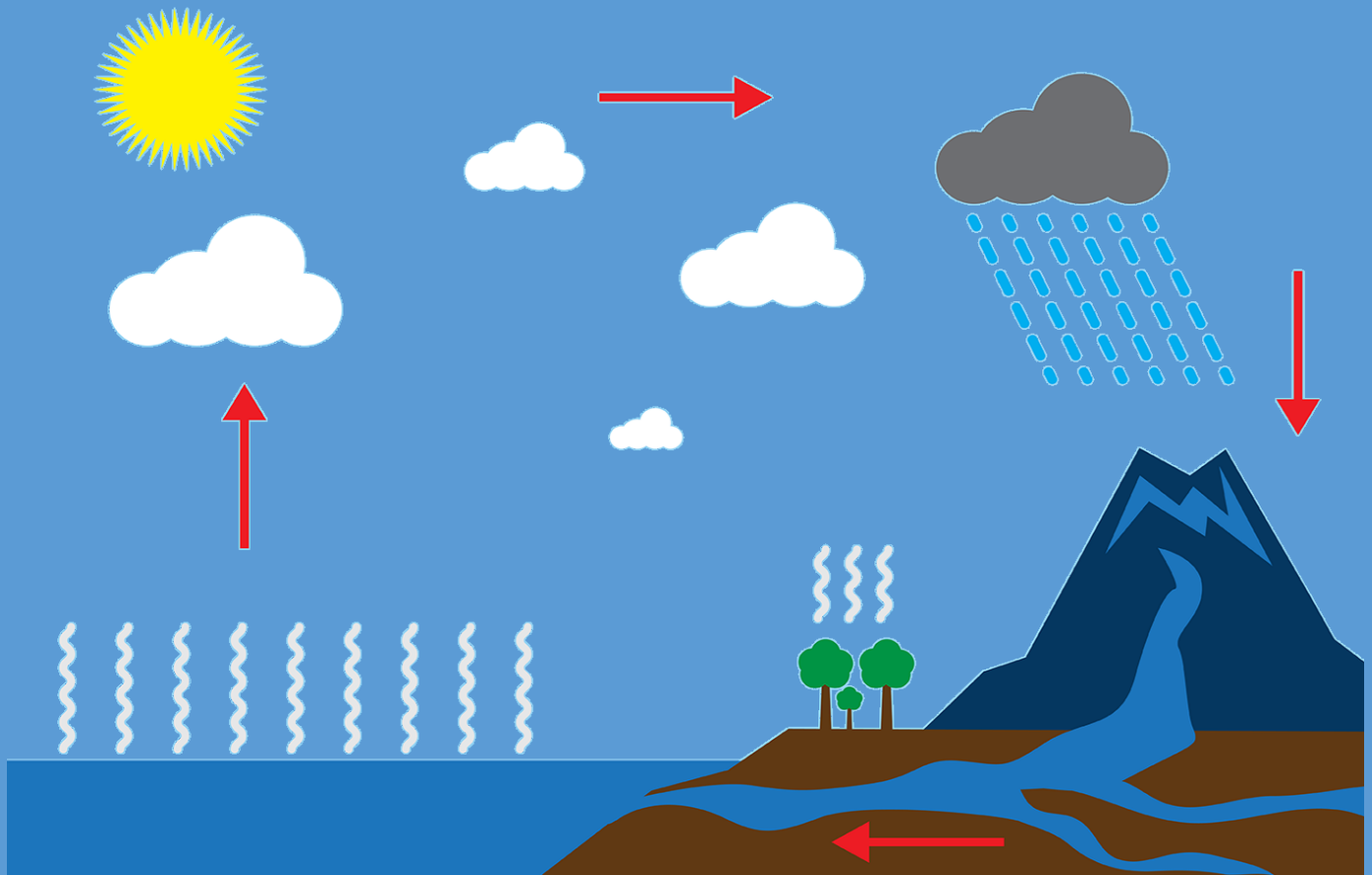
Label the following processes: (4 Marks)

- Evaporation
- Condensation
- Precipitation

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- Runoff



3. The teacher has provided you with a tomato plants that appears to have stunted growth and yellowing leaves. Using science knowledge:-
- a) Identify two possible causes of the problem. (2 Marks)

- b) Suggest two solutions to address the problem. (2 Marks)

4. **Simple Machines** (10 Marks)
- a) Draw and label a diagram of a lever. (5 Marks)

b) State two advantages of using simple machines in daily life. (2 Marks)

c) Give three example of a simple machine used at home. (3 Mark)

5. Test and classify the following specimens provided into acids, neutral or bases: (6 Marks)

Specimen	Base	Neutral	Acid
a) solution A			
b) Solution B			
c) Solution C			